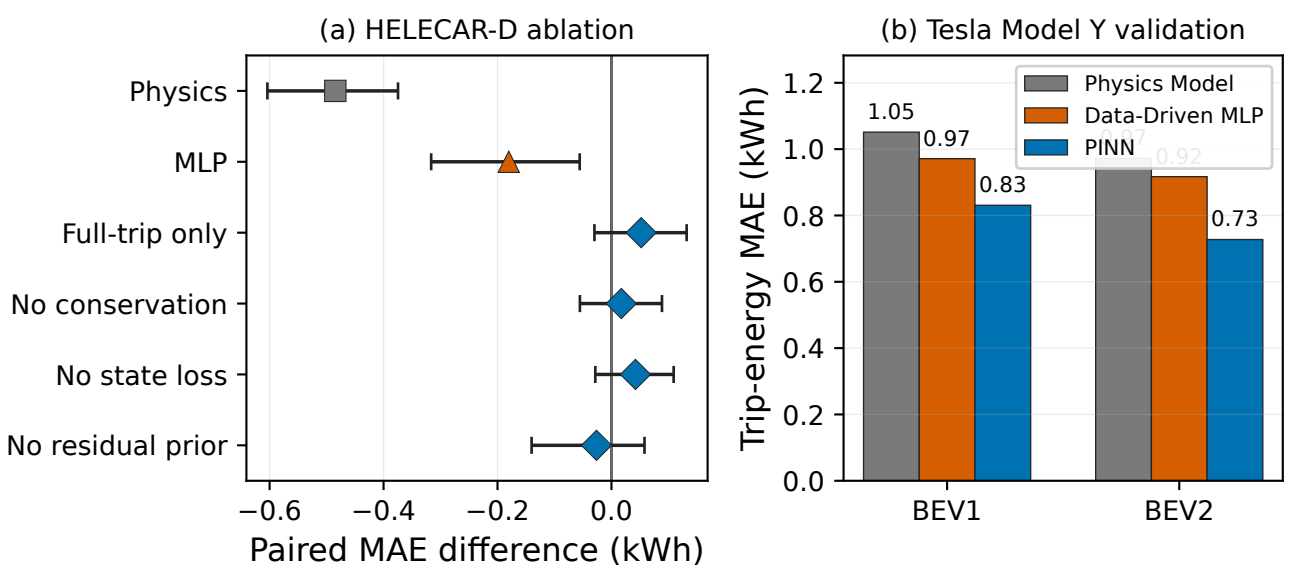
(a) HELECAR-D ablation
Physics
MLP
Full-trip only
No conservation
No state loss
No residual prior
−0.6
−0.4
−0.2
0.0
Paired MAE difference (kWh)
(b) Tesla Model Y validation
Physics Model
Data-Driven MLP
PINN
Trip-energy MAE (kWh)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.05
0.97
0.83
0.97
0.92
0.73
BEV1
BEV2